MATH 101 Final Exam - Practice 3

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Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10.

(*a*) $\log(0.05) + \log(200000)$

(*b*) $\dfrac{\log_8(125)}{\log_8(25)}$

2. Which pair satisfies h(x) = f(g(x))?

$h(x) = (5x+5)^2 + 6$

A. $f(x) = 5x^2 + 6,\ g(x) = x + 5$

B. $f(x) = 5x + 5,\ g(x) = x^2 + 6$

C. $f(x) = x^2 + 5,\ g(x) = 5x + 6$

D. $f(x) = x^2 + 6,\ g(x) = 5x + 5$

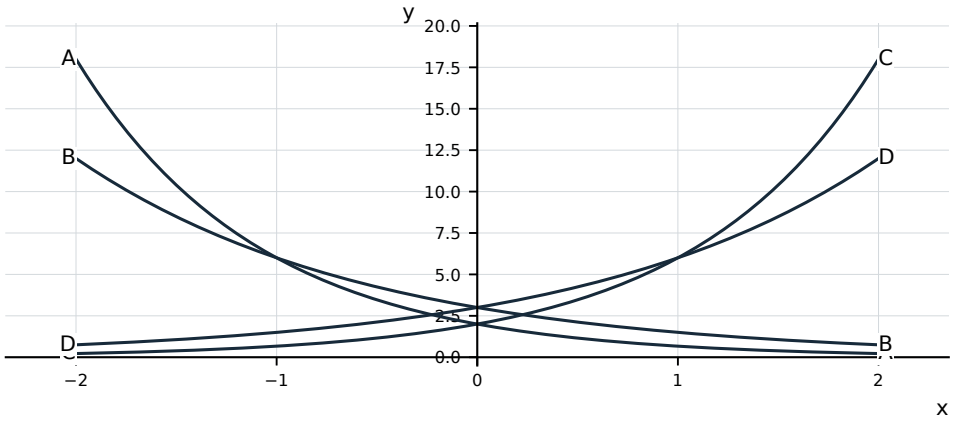
3. Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $2(3)^x$

(*b*) $2(\frac{1}{3})^x$

(*c*) $3(2)^x$

(*d*) $3(\frac{1}{2})^x$



4. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain.

$f(x) = 3\sqrt{x+16} - 12$

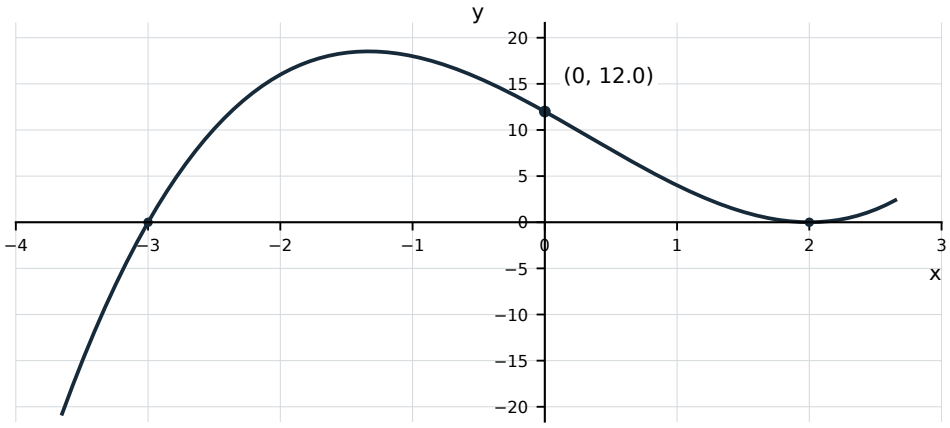
5. Find all zeros: (a) list Rational Root Theorem candidates, then test only -1 and 1; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros.

$$p(x) = x^3 + 3x^2 - 2$$

6. Solve exactly.

$$8^{x-3} = 4^{x+4}$$

7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



8. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain.

$$f(x) = \log_4(x - 4) + 3$$

9. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts.

$$f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$$

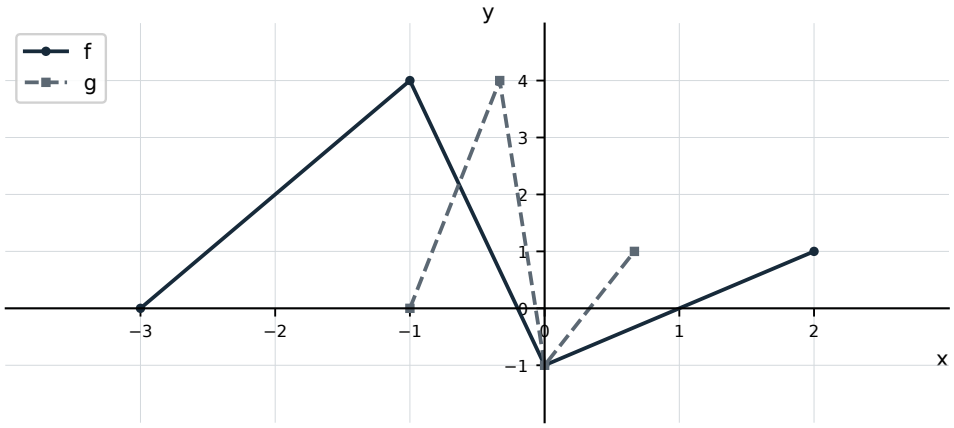
10. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(3x)$

B. $g(x) = f(x - 3)$

C. $g(x) = f(x/3)$

D. $g(x) = 3f(x)$



Student answer sheet

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Question 1

(a)	5 pts
(b)	5 pts

Question 2

Choice	10 pts

Question 3

(a)	3 pts
(b)	2 pts
(c)	3 pts
(d)	2 pts

Question 4

(a) Domain	2 pts
(a) Range	2 pts
(b) x-intercept	1 pts
(b) y-intercept	1 pts
(c) Infinite end	3 pts
(c) Finite end	1 pts

Question 5

(a) Candidates/root	2 pts
(b) Quotient	4 pts
(c) Other zeros	4 pts

Student answer sheet

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Question 6

Solution	10 pts

Question 7

Polynomial	10 pts

Question 8

(a) Domain	2 pts
(a) Range	2 pts
(b) Asymptote	1 pts
(b) x-intercept	1 pts
(c) Left end	2 pts
(c) Right end	2 pts

Question 9

(a) Domain	2 pts
(b) Hole	2 pts
(b) Vertical	1 pts
(b) Horizontal	1 pts
(c) x-intercept	2 pts
(c) y-intercept	2 pts

Question 10

Choice	10 pts

Answer key

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Question 1

(a)	4	5 pts
(b)	$\frac{3}{2}$	5 pts

Question 2

Choice	D	10 pts

Question 3

(a)	C	3 pts
(b)	A	2 pts
(c)	D	3 pts
(d)	B	2 pts

Question 4

(a) Domain	$[-16, \infty)$	2 pts
(a) Range	$[-12, \infty)$	2 pts
(b) x-intercept	$(0,\ 0)$	1 pts
(b) y-intercept	$(0,\ 0)$	1 pts
(c) Infinite end	$x \to \infty :\ f(x) \to \infty$	3 pts
(c) Finite end	$x \to -16^{+} :\ f(x) \to -12$	1 pts

Question 5

(a) Candidates/root	$\{-2, -1, 1, 2\};\ x = -1$	2 pts
(b) Quotient	$x^2 + 2x - 2;\ R = 0$	4 pts
(c) Other zeros	$x = -1 + \sqrt{3},\ \ -\sqrt{3} - 1$	4 pts

Answer key

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Question 6

Solution	$x = 17$	10 pts

Question 7

Polynomial	$1(x+3)(x-2)^2$	10 pts

Question 8

(a) Domain	$(4, \infty)$	2 pts
(a) Range	$(-\infty, \infty)$	2 pts
(b) Asymptote	$x = 4$	1 pts
(b) x-intercept	$(\frac{257}{64},\ 0)$	1 pts
(c) Left end	$x \to 4^+ :\ f(x) \to -\infty$	2 pts
(c) Right end	$x \to \infty :\ f(x) \to \infty$	2 pts

Question 9

(a) Domain	$\mathbb{R} \setminus \{-4, 7\}$	2 pts
(b) Hole	$(-4,\ -\frac{1}{11})$	2 pts
(b) Vertical	$x = 7$	1 pts
(b) Horizontal	$y = 1$	1 pts
(c) x-intercept	$(-5,\ 0)$	2 pts
(c) y-intercept	$(0,\ -\frac{5}{7})$	2 pts

Question 10

Choice	A	10 pts